

Equivalent mechanical model for liquid sloshing in irregular annular cylindrical containers of liquid metal reactors

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ARTICLE INFO

Keywords:

Equivalent mechanical model
Fluid-structure interaction (FSI)
Liquid metal reactor
Finite element method

ABSTRACT

The dynamic response of the liquid sloshing in the reactor vessel is an important issue for seismic designs of liquid metal reactors (LMR). In engineering design, equivalent mechanical models are generally used to evaluate the response of liquid containers. However, several equipment support cylinders are arranged in the reactor vessel forming an annular cylindrical container with evenly distributed cylinders. The equivalent model of such containers is absent. Therefore, we developed an equivalent mechanical model for liquid sloshing in the irregular annular region of LMR. Using the finite element method, parameters of the equivalent mechanical model were evaluated. Moreover, calculation formulas for parameters of the equivalent model were obtained. The calculations were compared with the experiments in the reference, achieving a good agreement. This study can provide parameters for a simplified model of liquid sloshing in similar containers and provide a reference for the seismic design of LMR.

1. Introduction

Liquid metal reactors (LMR), represented by lead-cooled fast reactors and sodium-cooled fast reactors, are one of the reactor types currently attracting the most attention from researchers because of their high thermal conductivity and excellent breeder characteristics (Alemberti et al., 2014; Aoto et al., 2014; Rehm, 2023). In comparison to pressurized water reactors, liquid metal reactors have several adjacent equipment support cylinders in the reactor vessel to support the steam generator, heat exchanger, and other equipment. These equipment support cylinders are bottom-fixed and closely arranged. Under seismic excitation, liquid sloshing in the reactor vessel can introduce additional loads on these equipment support cylinders and vessel walls, which in serious cases may compromise the structural integrity. The sloshing behavior of the LMR vessel under seismic excitation needs to be investigated comprehensively as it is critical to the safety and development of this type of reactor.

Liquid sloshing is related to the safety of the storage vessel, so the liquid sloshing problem was widely studied in the fields of nuclear engineering (Sanapala et al., 2016), petrochemical (Yamauchi et al., 2006), ocean engineering (Liu et al., 2023), space vehicles (Abramson, 1966) and civil engineering (Ibrahim et al., 2001). For liquid storage

containers with simple geometry, the theoretical solution of liquid sloshing with small amplitude can be obtained through boundary condition derivation. Based on the linear potential flow theory, Bauer (1960) and Aslam et al. (1979) derived a linear theoretical solution for liquid sloshing in an annular tank and verified the theoretical results through experiments. Abramson (1966) summarized the liquid sloshing behaviors in rectangular, cylindrical, and ring sector tanks analyzed by applying the velocity potential theory, and obtained theoretical solutions for sloshing frequency, wave height, and dynamic water pressure. Nevertheless, it is difficult to obtain the theoretical solution of liquid sloshing in this approach for some containers with complex geometry, which can only be analyzed by methods of numerical simulation or equivalent models.

Many researchers have carried out work on the numerical study of liquid sloshing. Numerical methods such as the boundary element method (BEM) (Dutta and Laha, 2000), finite difference method (FDM) (Chen and Nokes, 2005) finite element method (FEM) (Cho and Lee, 2004), the volume of fluid (VOF) method (Gómez-Goni et al., 2013) and smoothed particle hydrodynamics (SPH) method (Shao et al., 2012) have been applied. In many studies, the sloshing behavior in containers has been comprehensively analyzed using FLUID80 elements in ANSYS software (Jaiswal et al., 2008; Manser et al., 2008). Additionally,

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<https://doi.org/10.1016/j.anucene.2024.110425>

Received 12 December 2022; Received in revised form 27 July 2023; Accepted 10 February 2024

Available online 15 February 2024

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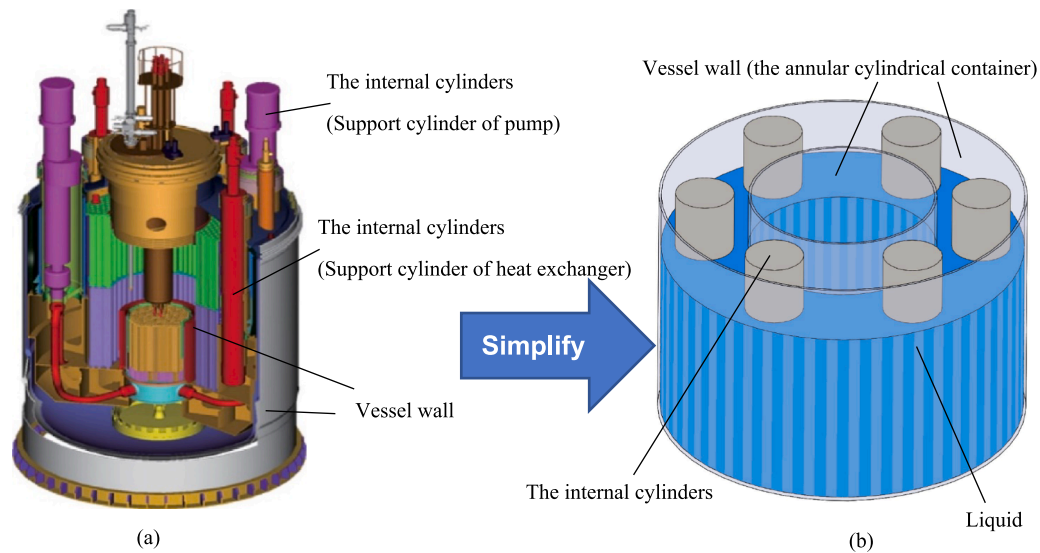


Fig. 1. Schematic diagram of the model in this paper (a) The reactor vessel of LMR (Lu et al., (2018)). (b) Simplified model of the reactor vessel.

numerical studies of liquid sloshing by other methods were conducted. For the passive containment tank of AP1000, Lu (2015) developed a finite element model using the FLUID30 and FLUID80 elements in ANSYS software and discussed the merits and demerits of the two classical acoustic fluid elements. Zhao et al. (2016) used the Arbitrary Lagrangian-Eulerian (ALE) method in the explicit dynamic code of LS-DYNA to analyze the liquid sloshing of irregular annular tanks in AP1000 nuclear power plants under seismic conditions and evaluated the dynamic response of different water level heights to the plant. Their research results provide a good reference for the seismic design of AP1000 and CAP1400 nuclear power plants. Using the coupled acoustics-structure method (CAS) and coupled Lagrangian Eulerian (CEL) in the commercial software ABAQUS, Rawat et al. (2019) analyzed the response of the liquid sloshing in a rigid container with small amplitude and compared the differences between the two numerical methods. Jeltsov et al. (2018) conducted a CFD numerical study of liquid sloshing under seismic conditions for the irregular liquid region between the two steam generators of a lead-cooled fast reactor. They innovatively proposed the addition of a baffle in the irregular liquid region as a mitigation measure for seismic sloshing and evaluated its effectiveness, which provides an important reference for the seismic design of lead-cooled fast reactors. By using CFD simulations in the code OpenFOAM and related experimental work, the liquid sloshing behavior of water and liquid metal was studied (Myrillas, 2016). The study pointed out that the use of water instead of liquid metal is an effective approach to evaluating sloshing behavior. Using water to simulate liquid sodium and simplifying the main pump and heat exchanger in a sodium-cooled fast reactor as a cylinder, Lu et al. (2018) investigated the dynamic characteristics of liquid sloshing in the reactor vessel by experiments and the VOF method, while obtaining a correction coefficient for calculating the sloshing frequency using a theoretical formulation. Using the SPH method, Jo et al. (2021) simulated the liquid sloshing phenomenon that may occur in the event of a core damage accident (CDA) in the sodium-cooled fast reactor and demonstrated the high accuracy of the simulation by comparing it with experiments. For the sloshing behavior of the container under longitudinal and oblique excitation, Sanapala et al. (2022) carried out a numerical study under resonance conditions whose results contribute to the design of liquid storage containers for nuclear engineering.

When evaluating the impact of liquid sloshing on liquid storage containers, the cost of using numerical methods for detailed modeling calculations is too high, making it difficult for engineers to use this method for early-stage engineering assessment. Moreover, structural

engineers primarily concentrate on the structural impact of the fluid rather than the fluid flow itself. Consequently, it is customary within the engineering field to adopt a simplified approach employing equivalent mechanical models to conveniently evaluate the repercussions of liquid sloshing on containers (API, 2020; AWWA, 2011). In the equivalent mechanical model of liquid sloshing, the pressure exerted by the fluid on the walls is simplified into two parts: the impulsive mass and the convective mass (Housner, 1963). The impulsive mass is rigidly connected to the container walls, while the convective mass is connected to the walls through a spring, with the frequency of the spring representing the sloshing frequency. Bauer (1960) obtained the theoretical solution for the dynamic pressure of the fluid in the annular cylindrical container and obtained the parameters of the equivalent mechanical model based on the conservation of kinetic energy and momentum. Li and Wang (2012) proposed a complementary solution of an equivalent mechanical model for a rectangular tank using the linear theory of potential flow for refining Housner's model. An equivalent mechanical model for studying the dynamic water pressure in a tank with baffles was developed and the effect of baffle properties on the parameters of the equivalent model was investigated (Ebrahimian et al., 2014). Based on Housner's model and the finite element method, Liu et al. (2015) developed an equivalent mechanical model for the elevated passive containment tank in AP1000 to study the dynamic pressure loads on the tank and investigated the effects of geometric parameters (inner and outer radius ratio, inclination angle of the tank, and the height) on the parameters of the equivalent model. In response to flexible cylindrical concrete liquid storage containers, Moradi et al. (2018) developed an equivalent mechanical model to assess the response of the containers under seismic conditions. By comparing it with the recommended values in ACI 350.3-06, the model was found to have good accuracy in predicting base shear and overturning moment. Li and Gou (2018) discussed the influence of modeling methods for equivalent mechanical models on the results, and modified the conventional formulas to achieve better results. Sun et al. (2019) established an equivalent mechanical model for linear liquid sloshing based on potential flow theory, specifically for cylindrical liquid storage containers with baffles. This model facilitates the convenient assessment of dynamic responses in complex fluid-structure interaction systems. In response to rectangular water tanks containing internal porous media, Tsao et al. (2021) conducted experimental research and proposed a modified equivalent mechanical model that takes into account nonlinear damping. Hu et al. (2022) reexamined the position of the sloshing mass in the equivalent mechanical model for cylindrical liquid storage containers. Based on formula derivations, they proposed new analytical

formulas. In addressing the liquid sloshing issue in spacecraft, Mingle et al. (2022) utilized the Moving Pulsation Ball Model (MPBM) to establish an equivalent mechanical model for nonlinear rotational sloshing. Through experimental comparisons of the sloshing forces and moments, they validated the effectiveness of the model. Hosseini and Beskhyroun (2023) comprehensively summarized the development and application of equivalent mechanical models for liquid storage containers over the past few decades. The equivalent mechanical model provides a highly approximate representation of the structural effects of the actual fluid, including the overall force and moment exerted by the liquid on the structure. The use of equivalent mechanical models can simplify the calculations involved in fluid–structure interaction and improve efficiency for structural analysis.

The reactor vessel of a liquid metal reactor with equipment support cylinders (as shown in Fig. 1(a)) can be approximated as an annular cylindrical container with evenly distributed cylinders. And the simplified model is shown in Fig. 1(b). Although many studies have been performed by researchers on equivalent mechanical models for liquid sloshing in different shaped containers, there are few equivalent mechanical models of such containers. Moreover, the presence of the cylinder affects the dynamic behavior of the liquid and leads to changes in the parameters of the equivalent mechanical model, which must be considered in the analysis and which are currently lacking in engineering.

In this study, an equivalent mechanical model was developed based on Housner's model, which applies to studying the liquid sloshing response of annular cylindrical containers with evenly distributed cylinders. The effect of the geometric parameters of the cylinder on the sloshing response of the container was investigated by using the new high-order acoustic fluid element FLUID220 in the finite element software ANSYS and evaluating the parameters of the equivalent mechanical model. By fitting the calculated results, formulas for the parameters of an equivalent mechanical model of an annular cylindrical container with evenly distributed cylinders were obtained and compared with the experimental results. The results of this study can provide parameters for a simplified model of liquid sloshing in a similar container, which can be used to estimate the fluid loads due to sloshing in the reactor vessel of LMR with reasonable accuracy as a reference for practical engineering construction.

2. Methods

2.1. Finite element method for acoustic fluid elements

The finite element method (FEM) using acoustic fluid elements is generally considered to be an effective method for studying the sloshing response of liquid storage containers. In this study, we evaluate the parameters of the equivalent mechanical model using the FEM with acoustic fluid elements to compare the dynamic characteristics of the container.

A structural control equation can be derived from the finite element formulation which is described as:

$$M\ddot{u} + C\dot{u} + Ku = F \quad (1)$$

where M , C , and K are the mass, damping, and stiffness matrices of the structure respectively. u is the displacement vector of the structure and F is the sum of the external forces applied. For the fluid–structure interaction problem, the controlling equation of the coupled fluid–structure interface is required, so that F in Eq.(1) can be generalized as:

$$F = F_p + F' \quad (2)$$

where F_p is the integral of the pressure on the fluid at the fluid–structure interaction interface, and F' is the external force.

It is assumed that the fluid is incompressible and inviscid. Thus, the Navier-Stokes equations and continuity equations of the fluid can be

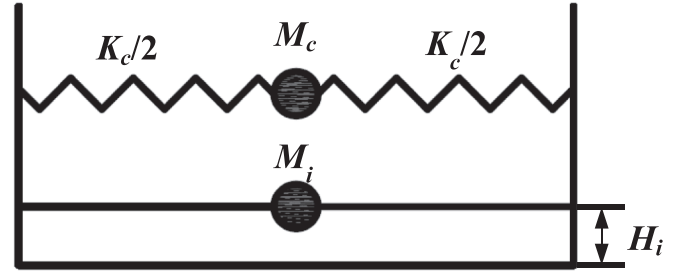


Fig. 2. Housner's equivalent mechanical model.

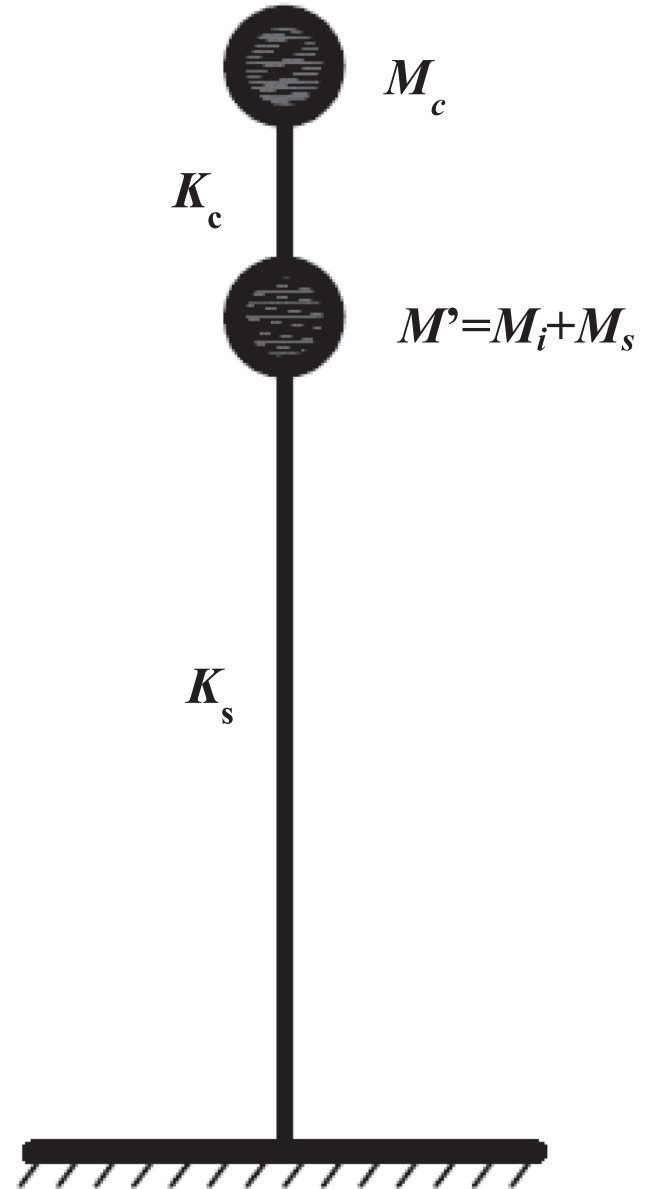


Fig. 3. The equivalent model in this study.

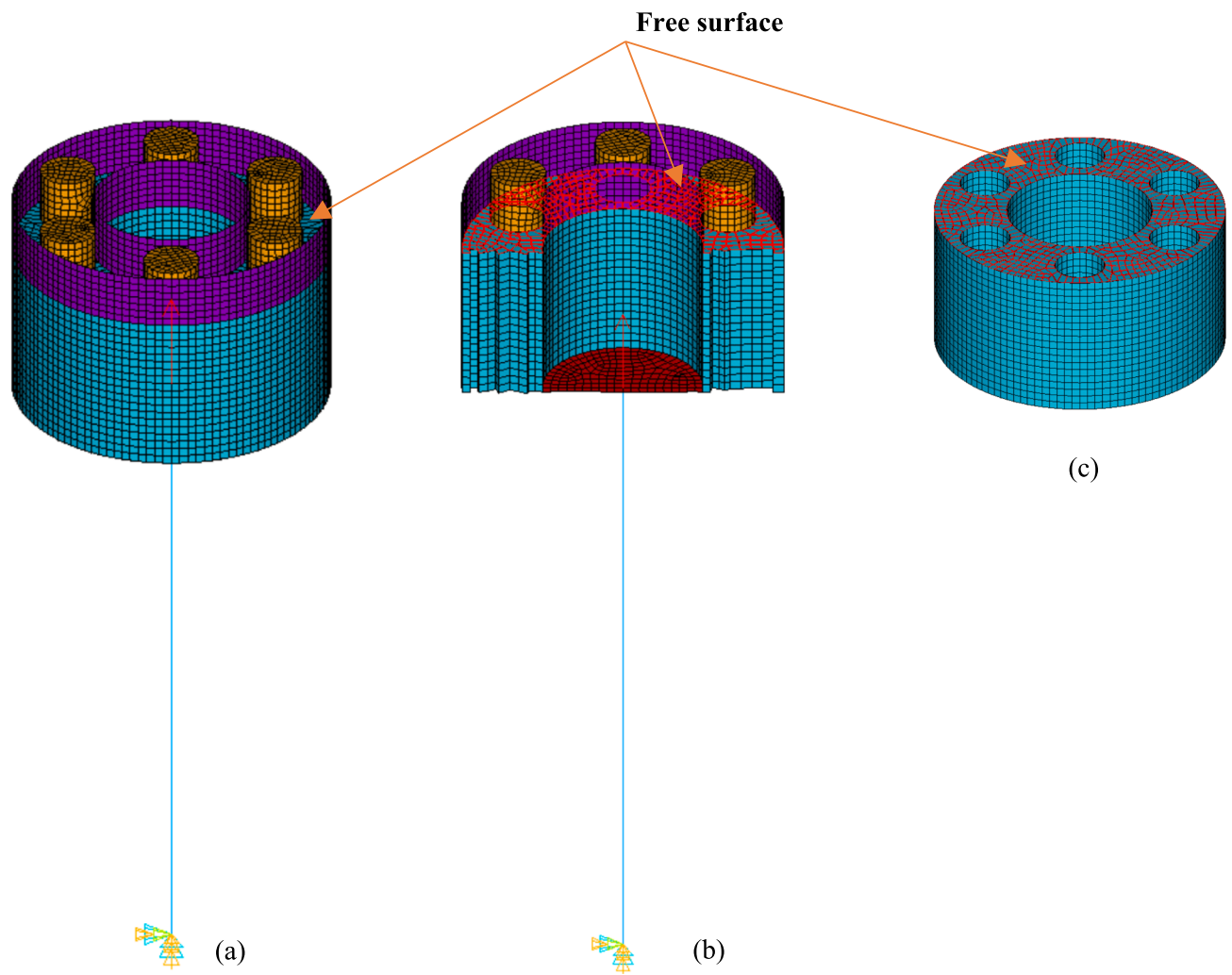


Fig. 4. The finite element model in ANSYS (a) Complete model. (b) Sectional view. (c) Liquid elements.

Table 1

The geometry parameters of the finite element model.

Geometry	Dimension (m)		
	Height	Diameter	Thickness
Annular cylindrical container	10	7.5/15	0.065
Cylinders	10	1.875	0.040
Water	7.5	7.5/15	–
Supporting beam	30*	15	0.065

*The height here was obtained by the relevant calculation in Section 3.2.

simplified to a pressure fluctuation equation, which is given as:

$$\frac{1}{c} \frac{\partial^2 p}{\partial t^2} - \nabla^2 p = 0 \quad (3)$$

where c is the velocity of sound in the fluid, p is the pressure, and t is the time. We can get the eigenvalues of the fluid–structure interaction problem by using the asymmetric algorithm in the modal analysis of ANSYS.

2.2. Equivalent mechanical model

Based on the work of Bauer (1960) and Housner (1963), the dynamic behavior in a water container can be simplified to an equivalent mechanical model that consists of mass points and corresponding springs as shown in Fig. 2. Presently, transforming irregularly shaped containers into similar equivalent mechanical models is a common method to study the sloshing of liquid storage containers.

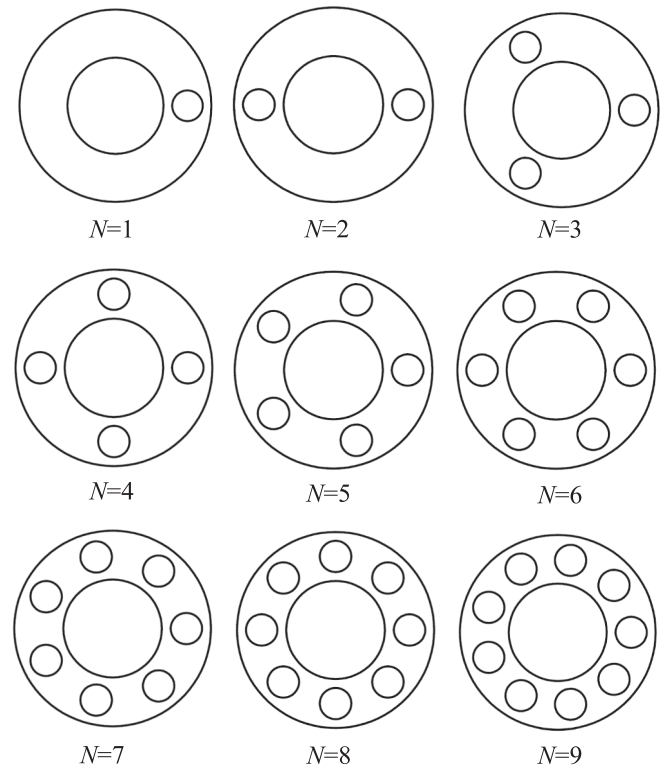
Based on Housner's model, Liu et al. (2015) developed an equivalent mechanical model for an elevated container with an irregular shape and proposed an equation to calculate the parameters of the mechanical model using the model's natural frequency. The detailed derivation of this equation can be found in the work of Liu et al. (2015).

$$M' = -\frac{fdM}{2df} \quad (4)$$

where M' is the sum of the impulsive mass M_i and the equivalent structural mass, and f is the natural frequency of the model. M' can be obtained using this equation by increasing the mass of the model (achieved by slightly increasing the thickness of the walls) while changing the natural frequency of the model. The impulsive mass M_i can be obtained by M' minus the equivalent structural mass. And the convective mass M_c can be approximated by subtracting M_i from the mass of water. The limitation of using Eq.(4) to calculate the impulsive mass M_i is the need to ensure that the vibration mode corresponding to the natural

frequency f used in the calculation is a beam vibration mode.

In this study, we developed an equivalent mechanical model of the cylindrical container with evenly distributed cylinders, which is shown in Fig. 3. This model was simplified to a two-degree-of-freedom system consisting of a supporting beam, an impulsive mass, a convective mass, and a container mass. The container mass and the impulsive mass are connected directly to the beam, while the convective mass is connected to the container wall by a spring with stiffness K_c . In general, the parameters of the equivalent mechanical model can be obtained by conservation of kinetic energy and conservation of momentum if an expression for the dynamic pressure of the fluid in the container is available. However, due to the complex boundary conditions of the container, it is difficult to obtain the theoretical expressions for the fluid dynamic pressure and the general method cannot be used. Therefore, the

**Fig. 5.** Schematic diagram of the model with different numbers of cylinders.**Table 2**

The material properties of the finite element model.

Geometry	Element types	Material	Material properties			
			Density (kg/m ³)	Elastic module (GPa)	Poisson ratio	acoustic velocity (m/s)
container and cylinders	SHELL281	Steel	7850	201	0.3	–
Water	FLUID220	Water	1000	–	–	1400
Supporting structure	BEAM188	–	7.85	201	0.3	–

Table 3The comparison of M_i/M by Bauer and this study.

Beam height (H_s/R_{out})	$H_w/R_{out} = 0.5$			$H_w/R_{out} = 0.75$			$H_w/R_{out} = 1$		
	This study	Bauer	Errors (%)	This study	Bauer	Errors (%)	This study	Bauer	Errors (%)
3.75 m (0.5)	0.5297	0.4054	23.46	0.6458	0.4978	22.92	0.7221	0.5777	19.99
7.5 m (1.0)	0.4516		10.24	0.5602		11.14	0.6373		9.35
15 m (2.0)	0.3918		3.46	0.5215		4.54	0.5926		2.52
30 m (4.0)	0.4097		1.05	0.4990		0.24	0.5760		0.29
60 m (8.0)	0.3986		1.72	0.5004		0.52	0.5701		1.34

where H_s is the height of the supporting beam, H_w is the height of liquid, R_{out} is the outer radius of the annular cylindrical container.

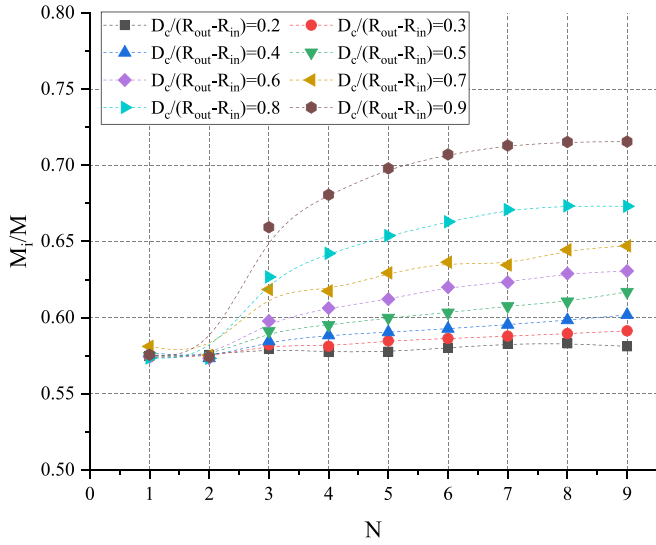


Fig. 6. Variation of impulsive mass with the number of cylinders N .

finite element method of ANSYS Mechanical APDL and Eq.(4) was used to obtain the relevant parameters.

3. Finite element calculation and verification of the annular cylindrical container with evenly distributed cylinders

3.1. The finite element model

The modal analysis in ANSYS can be used to obtain the natural frequency of an arbitrarily shaped container. However, the reactor vessel of the LMR does not belong to a typically elevated container, and it is difficult to obtain the beam vibration mode if the finite element model is established directly, which means that Eq.(4) cannot be used. Therefore, we used ANSYS to establish the finite element models with a supporting beam, which is shown in Fig. 4. It is worth noting that the supporting beam here is not part of the real structure and was established only to easily obtain the beam vibration mode, so its density was defined to be small enough to eliminate the influence on the real results.

The acoustic fluid element FLUID220 was used to simulate the liquid in the container. The shell element SHELL281 was used to simulate the wall of the container, and the BEAM188 element was used to simulate the supporting structure. The geometric parameters of the finite element model are shown in Table 1, and the material properties are shown in Table 2. The bottom of the supporting beam was fixed, the walls of the container and the cylinder in contact with liquid were set as the surface of fluid–structure interaction (FSI interface), the liquid–air contact surface was set as the free surface, and a gravitational acceleration of 9.81 m/s^2 was applied to the model in the vertical direction.

3.2. Validation of the equivalent mechanical model

To evaluate the validity of the equivalent mechanical model in this study, finite element models were developed for annular cylindrical containers with different lengths of supporting beams. The impulsive mass was then computed using Eq. (4). The calculated results are compared with the theoretical results for the impulsive mass of the annular cylindrical container proposed by Bauer (1960). The comparison is shown in Table 3.

The comparison in Table 3 shows that the results of the equivalent mechanical model are very close to Bauer's theoretical results with a maximum error of no more than 0.3 % at the height of the support beam of 30 m ($H_s/R_{out} = 4.0$). Thus, in this study, we set the height of the

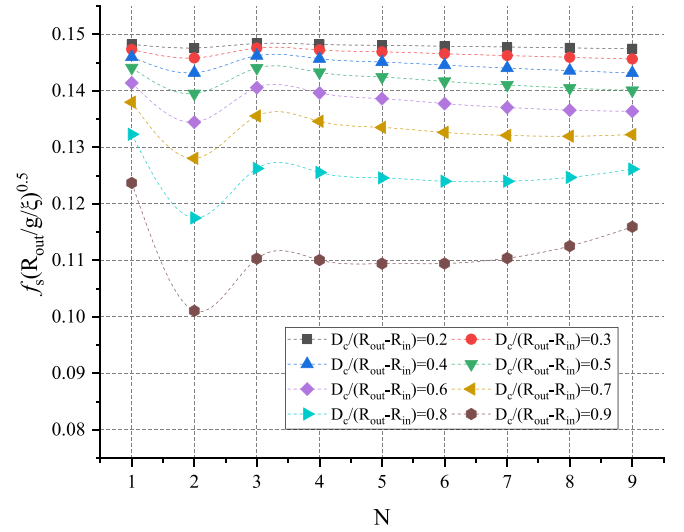


Fig. 7. Variation of the parameter Π with the number of cylinders N .

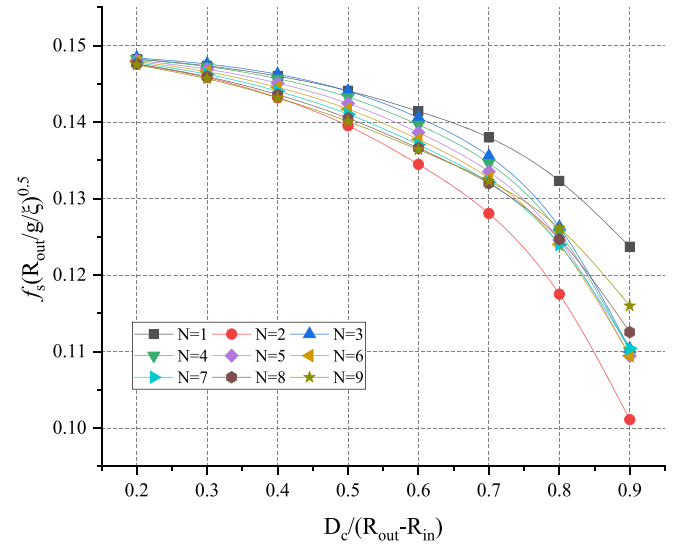


Fig. 8. Variation of the parameter Π with $D_c/(R_{out}-R_{in})$.

supporting beam for the equivalent mechanical model to 30 m. See (Fig. 5).

4. Effect of different geometric parameters on the equivalent mechanical model

Different types of liquid metal reactors contain different numbers and dimensions of cylinders in the reactor vessel. These differences may affect the sloshing characteristics. Therefore, it is necessary to investigate the effect of different geometrical parameters (number and dimensions of the internal cylinders) on the equivalent mechanical model. The annular cylindrical containers with evenly distributed cylinders inside were designed, which are shown in Fig. 4. A non-dimensional parameter $D_c/(R_{out}-R_{in})$ is defined to characterize the diameter of these cylinders. Here D_c is the diameter of the cylinder, R_{out} and R_{in} is the outer and inner radius of the annular cylindrical container respectively. In this study, the number of cylinders N ranged from 1 to 9, and the non-dimensional parameter $D_c/(R_{out}-R_{in})$ ranged from 0.2 to 0.9.

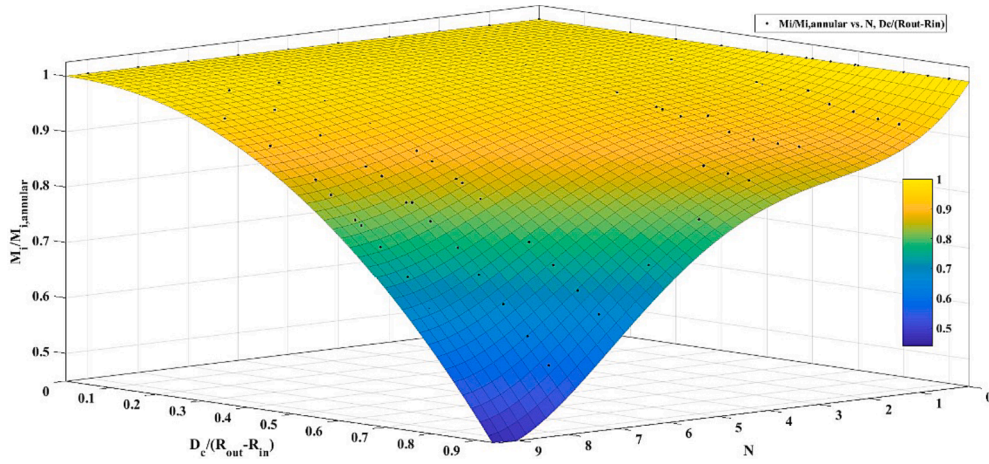


Fig. 9. Fitting results of the impulsive mass.

4.1. Impulsive mass

The variation of the non-dimensional impulsive mass M_i/M with different N and $D_c/(R_{\text{out}}-R_{\text{in}})$ is illustrated in Fig. 6, where M is the mass of water.

From the curves of Fig. 6, it can be found that M_i/M is almost constant for the cases of $N = 1$ and 2. In the case of $N \geq 3$, M_i/M increases obviously with increasing N . It can be seen that M_i/M varies insignificantly with $D_c/(R_{\text{out}}-R_{\text{in}})$ for the cases of $N = 1$ and 2. For the other cases, M_i/M shows an increase with $D_c/(R_{\text{out}}-R_{\text{in}})$ with a similar exponential trend.

These results illustrate that the impulsive mass increases with the number and dimensions of the cylinders, and this trend is more pronounced for larger cylinder dimensions ($D_c/(R_{\text{out}}-R_{\text{in}}) \geq 0.5$) or a larger number of cylinders ($N > 3$). In the cases where the number of cylinders N is 1 and 2, the variations in cylinder size have minimal influence on the impulsive mass. However, when the number of cylinders $N \geq 3$, there is a significant increase in the impulsive mass with the increasing dimension of the cylinders, and the magnitude of the increase gradually rises. Therefore, in the engineering design of liquid metal reactors, particular attention should be paid to situations where the number of internal equipment exceeds 3, and the dimensions of the devices should be designed appropriately.

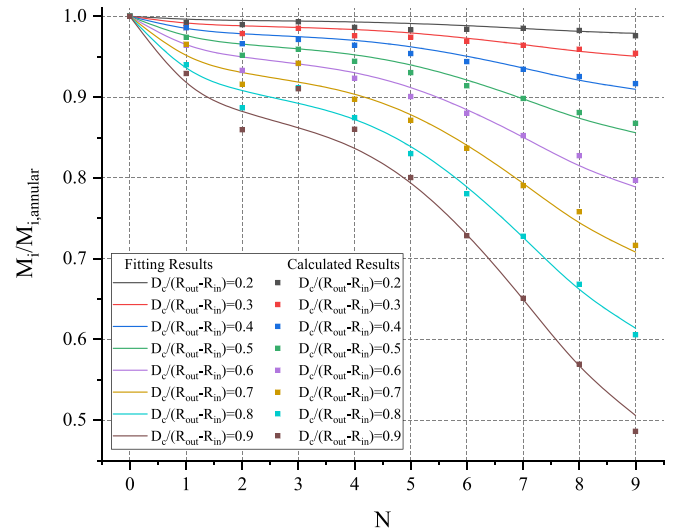
4.2. Sloshing frequency and convective mass

A non-dimensional parameter $\Pi = f_s(R_{\text{out}}/g/\xi)^{0.5}$ is defined to characterize the sloshing frequency, where f_s is the sloshing frequency of liquid and ξ is a constant which depends on $R_{\text{in}}/R_{\text{out}}$. The sloshing frequency can be obtained by using the finite element method of ANSYS. The variation of the non-dimensional parameter Π with N and $D_c/(R_{\text{out}}-R_{\text{in}})$ is shown in Fig. 7 and Fig. 8.

Furthermore, the sloshing frequency f_s in this paper is the frequency f_c of the convective spring in the equivalent mechanical model. The convective mass M_c can be approximated by subtracting the impulsive mass from the total mass of water. Therefore, the spring stiffness of the convective mass can be obtained by the following equation.

$$K_c = M_c(2\pi f_c)^2 \quad (5)$$

From Fig. 7, it can be found that as the number of cylinders N increases from zero, the non-dimensional parameter Π first decreases and appears as a minimum at $N = 2$. Subsequently, as N increases from 2 to 3, Π then increases and stabilizes after $N > 3$. This trend becomes more obvious as $D_c/(R_{\text{out}}-R_{\text{in}})$ increases. From Fig. 8, it can be found that the non-dimensional parameter Π decreases nonlinearly with increasing $D_c/(R_{\text{out}}-R_{\text{in}})$.

Fig. 10. Comparison between calculated and fitting results of impulsive mass (with N as the independent variable).

We found that the increase in the cylinder dimensions leads to the decrease in the sloshing frequency. In contrast, the change in the cylinder number leads to a significant change in the sloshing frequency only for $1 \leq N \leq 3$, which indicates that the effect of cylinder dimensions on the sloshing frequency is more significant than the effect of the number of cylinders.

In this study, there are two factors that affect the liquid sloshing frequency: the area of the free surface and the geometric shape of the free surface. Generally, after excluding other factors, the sloshing frequency increases as the area of the free surface decreases. When the characteristics of the cylinder change (dimensions and number), both the area and geometric shape of the free liquid surface also change, and these two factors together influence the sloshing frequency. In the case where the dimensions of the cylinder remain unchanged, when the number of cylinders N increases from 1 to 2, the area of the free surface decreases, but the sloshing frequency decreases significantly. This indicates that, in this case, the geometric shape of the free surface dominates the influence on the sloshing frequency, while the area has a secondary impact. When the number of cylinders N increases from 2 to 3, the sloshing frequency increases, indicating that the area of the free surface becomes the dominant factor affecting the sloshing frequency. As the number of cylinders continues to increase, the sloshing frequency remains relatively stable.

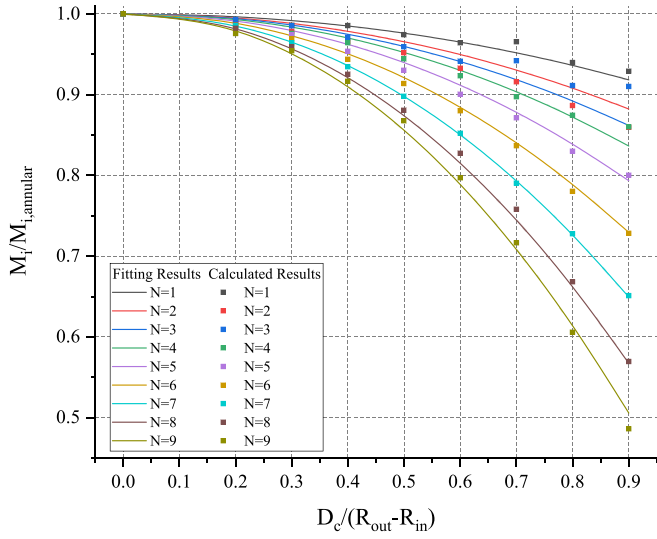


Fig. 11. Comparison between calculated and fitting results of impulsive mass (with $D_c/(R_{out}-R_{in})$ as the independent variable).

5. Formula for the parameters of the equivalent mechanical model

5.1. Description of the fitting approach

In the study of Bauer (1960), the frequency $f_{s,annular}$ of liquid sloshing in an annular cylindrical container can be written as:

$$f_{s,annular} = \frac{1}{2\pi} \sqrt{\frac{g\xi}{R_{out}} \tanh\left(\frac{\xi}{R_{out}} H_w\right)} \quad (6)$$

And the impulsive mass $M_{i,annular}$ of the annular cylindrical container can be expressed as:

$$M_{i,annular} = M \left\{ 1 - \sum_{n=1}^{\infty} \frac{\bar{A}_n \tanh(\xi H_w / R_{out})}{(1 - k^2)(\xi H_w / R_{out})} \left[\frac{2}{\pi} - k C_1(k\xi) \right] \right\} \quad (7)$$

$$\bar{A}_n = \frac{2[(2/\pi\xi) - k C_1(k\xi)]}{(2/\pi\xi)^2(\xi^2 - 1) + C_1^2(k\xi)(1 - k^2\xi^2)} \quad (8)$$

where R_{out} is the outer radius of the annular cylindrical container, H_w is

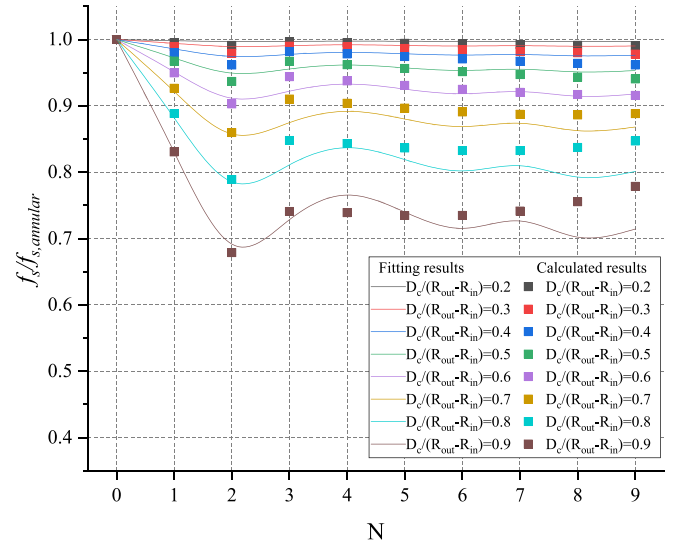


Fig. 13. Comparison between calculated and fitting results of sloshing frequency (with N as the independent variable).

the height of the liquid, M is the total mass of water, k is the ratio of the inner and outer radius, ξ and C_1 are constants that depend on k .

From Eqs.(6) to (8), it can be seen that the sloshing frequency and impulsive mass of the annular cylindrical container are related to the geometry of the container and the height of the liquid. To evaluate the effect of the internal cylinders on the above parameters, the sloshing frequency f_s and impulsive mass M_i of the model in this paper were divided by the sloshing frequency $f_{s,annular}$ and impulsive mass $M_{i,annular}$ of the annular cylindrical container with the same geometry, respectively. The ratios ($f_s / f_{s,annular}$ and $M_i / M_{i,annular}$) were taken as the dependent variables of the fitting function.

In this study, the function fitting was implemented using the fitting toolbox in MATLAB. Two parameters, R -Square and Sig. were applied to assess the fitting results. Here, R -square is the squared value of the correlation coefficient between the original data and the fitted data, and Sig. is the sum of squares of the errors of the fitted and original data. The value of R -square tends to be 1 and the value of Sig. tends to 0, indicating a better fit.

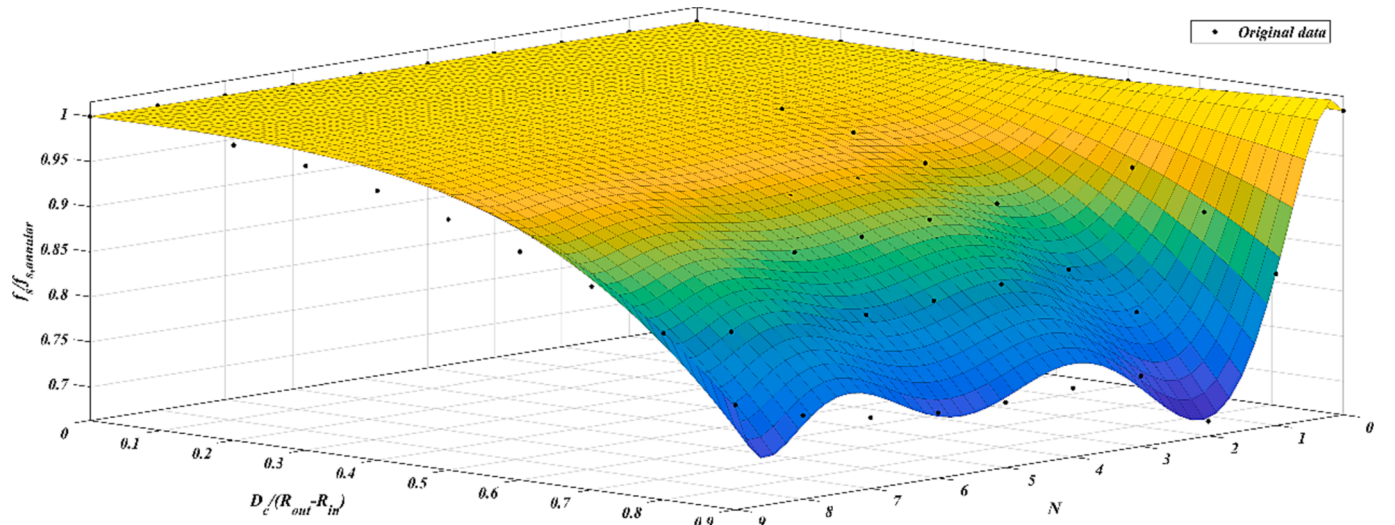


Fig. 12. Fitting results of the sloshing frequency.

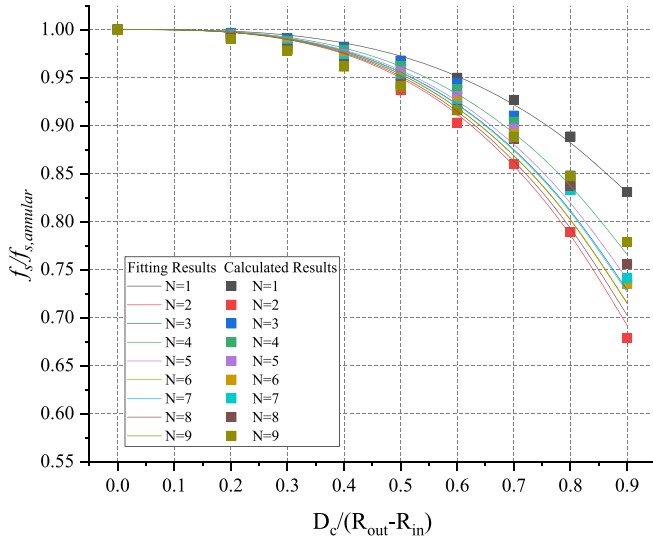


Fig. 14. Comparison between calculated and fitting results of sloshing frequency (with $D_c/(R_{out}-R_{in})$ as the independent variable).

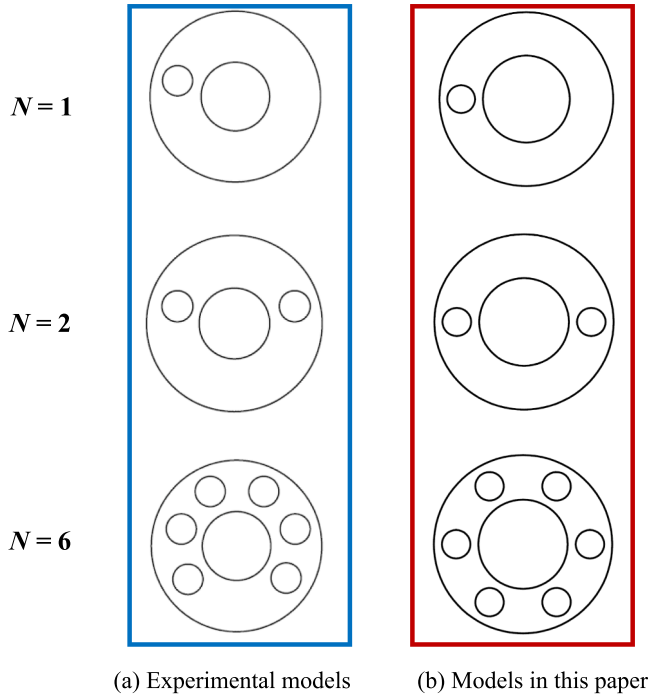


Fig. 15. The experimental models and the models in this paper. (a) Experimental models (Liu, 2018; Lu et al., 2018). (b) Models in this paper.

5.2. Formula of the impulsive mass

The data were fitted in three dimensions using a custom function, with the number of cylinders N and the dimension of the cylinders $D_c/(R_{out}-R_{in})$ as the independent variables of the fitting function and $M_i/M_{i,annular}$ as the dependent variable of the fitting function. The fitting function is:

$$M_i/M_{i,annular} = 1 - (\alpha_1 N^4 + \alpha_2 N^3 + \alpha_3 N^2 + \alpha_4 N) \cdot [D_c/(R_{out} - R_{in})]^\beta \quad (9)$$

where $\alpha_1, \alpha_2, \alpha_3, \alpha_4$, and β are coefficients to be determined.

By fitting the calculated data, the formula of the impulsive mass M_i for the annular cylindrical container with evenly distributed cylinders is

applicable to the different number of cylinders ($N = 0, 1, 2, \dots, 9$) and different cylinder dimensions ($0.2 \leq D_c/(R_{out}-R_{in}) \leq 0.9$) was obtained, which can be expressed as:

$$M_i = M_{i,annular} \cdot \left\{ 1 - (-0.0004524N^4 + 0.00889N^3 - 0.05187N^2 + 0.1452N) \cdot [D_c/(R_{out} - R_{in})]^{2.097} \right\} \quad (10)$$

The fitting result is shown in Fig. 9 with the R -Square of 0.9928 and Sig. of 0.01086. The comparison between the fitting results and the calculated data is shown in Fig. 10 and Fig. 11. The fitting results show a good agreement with the calculated results obtained using the equivalent mechanical model, especially in the case of a high number of cylinders ($N \geq 3$).

5.3. Formula of the sloshing frequency

Similarly, the data were fitted in three dimensions using a custom function, with the number of cylinders N and the dimension of the cylinders $D_c/(R_{out}-R_{in})$ as the independent variables of the fitting function and $f_s/f_{s,annular}$ as the dependent variable of the fitting function. According to our work, it is found that the best fit is achieved when the order of the polynomial is 7th order. Therefore, the fitting function is:

$$f_s/f_{s,annular} = 1 - (\gamma_1 N^7 + \gamma_2 N^6 + \gamma_3 N^5 + \gamma_4 N^4 + \gamma_5 N^3 + \gamma_6 N^2 + \gamma_7 N) \cdot [D_c/(R_{out} - R_{in})]^\delta \quad (11)$$

where $\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5, \gamma_6$ and δ are coefficients to be determined.

By fitting the calculated data, the formula of the sloshing frequency f_s for the annular cylindrical container with evenly distributed cylinders applicable to the different number of cylinders ($N = 0, 1, 2, \dots, 9$) and different cylinder dimensions ($0.2 \leq D_c/(R_{out}-R_{in}) \leq 0.9$) was obtained, which can be expressed as:

$$f_s = f_{s,annular} \cdot \left\{ 1 - (-6.978 \times 10^{-5} N^7 + 0.002229 N^6 - 0.02889 N^5 + 0.1862 N^4 - 0.6083 N^3 + 0.8684 N^2 - 0.1848 N) \cdot [D_c/(R_{out} - R_{in})]^{3.08} \right\} \quad (12)$$

The fitting result is shown in Fig. 12 with the R -Square of 0.9928 and Sig. of 0.01086. The comparison between the fitting results and the calculated data is shown in Fig. 13 and Fig. 14. The fitting results show a good agreement with the calculated results obtained using the equivalent mechanical model. Similar to the fitting results of the impulsive mass, the fitting of the sloshing frequency has a better agreement for the case with a larger number of cylinders ($N \geq 3$).

6. Validation of the formula

Liquid sloshing frequencies calculated by using Eq.(12) were compared with the experimental results (Liu, 2018; Lu et al., 2018) to verify the fitting formula. The geometry of the experimental model differs from that of the model in this study, so we only selected the results of the experimental model that is similar to the model studied in this paper for comparison, as shown in Fig. 15. Additionally, we performed the simulations for both experimental and evenly distributed of cylinders. We then compared the FEM values with experimental values and calculated values. The comparison results are presented in Fig. 16 and Table 4.

Based on Fig. 16, it is observed that the sloshing frequency of the equivalent mechanical model decreases significantly with an increase in the cylinder's quantity N from 0 to 2. However, as N continues to increase, the frequency stabilizes and shows only a slight variation.

It can be found from the comparison that the results calculated by fitting the formulas in the cases of $N = 1$ and 2 are in good agreement with the experimental results with an error of less than 4 % (The experimental relative uncertainty of frequency is about 3 %). In the case

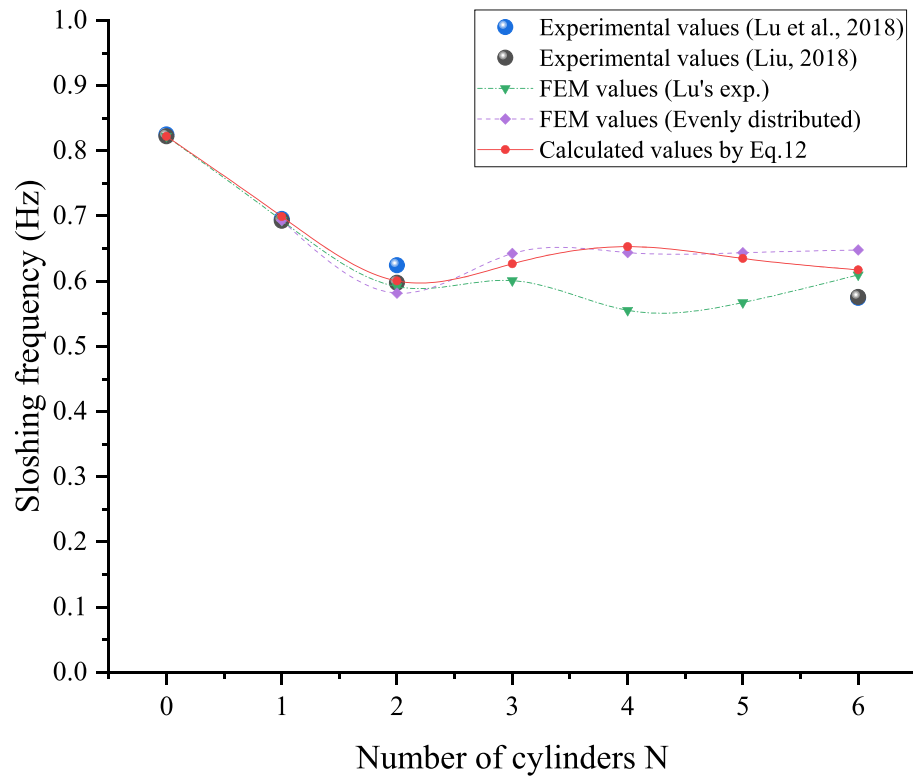


Fig. 16. Comparison of sloshing frequency: Experimental values, FEM values and Calculated values.

Table 4

Comparison of sloshing frequency: Experiment, FEM and Fitted formula.

Number of cylinders N	Sloshing frequency f_s (Hz)				
	Experimental values (Lu et al., 2018)	Experimental values (Liu, 2018)	FEM values (Lu's exp.)	FEM values (Evenly distributed)	Calculated values by Eq. 12
1	0.6954	0.6928	0.6933	0.6933	0.6991
2	0.6245	0.5975	0.5916	0.5817	0.6004
6	0.5744	0.5757	0.6097	0.6480	0.6173

of $N = 6$, the error in the calculated results is slightly larger. This should be attributed to the fact that the cylinders are not evenly distributed in the experimental work. Based on the results above, it can be concluded that the FEM simulations are trustworthy.

The purpose of this paper is to provide a fast calculation method for circular containers with inner cylinders. According to the results shown in the above figure, the fitting formula provided in this paper has an error within 4.7 % compared to the values obtained from FEM. Moreover, the error of the calculation results between the fitting formula (evenly distributed) and experiment (not evenly distributed) is within 7.3 %. The above results show the fitting equations are acceptable for the preliminary reactor design, proving that the fitting formula in this study can better reflect the influence of the internal cylinders on the parameters of the equivalent mechanical model.

7. Conclusions

In this study, an equivalent mechanical model consisting of the impulsive part and the convective part was developed, which was used to study the liquid sloshing response of an annular cylindrical container with evenly distributed cylinders. Using the finite element method, the parameters of the mechanical model were evaluated and represented graphically as a function of the container's geometric parameters, including the number and the diameter of the internal cylinders. The

formulas on the impulsive mass and sloshing frequency of the container were fitted by the calculation of the equivalent mechanical model. The detailed conclusions are presented below.

- 1) Finite element models established by using a new high-order acoustic-fluid element FLUID220 can well evaluate the dynamic response to liquid sloshing in the container.
- 2) The impulsive mass of the equivalent mechanical model increases with the number and dimensions of the cylinders. And this trend is more significant for larger cylinder dimensions ($D_c/(R_{out}-R_{in}) \geq 0.5$) or a larger number of cylinders ($N > 3$).
- 3) The sloshing frequency of the equivalent mechanical model decreases remarkably as the cylinder's quantity N increases from 0 to 2. As N continues to increase, the frequency remains a relatively stable value with a small variation.
- 4) By comparing with the experimental results, the validity of the formula for calculating the parameters of the equivalent mechanical model was verified. The formulas can calculate the parameters of the equivalent mechanical model and provide important input parameters for similar mechanical models.

CRediT authorship contribution statement

Yuxuan Zhu: Methodology, Software, Validation, Writing – original

draft. **Daogang Lu:** Resources, Supervision, Funding acquisition, Writing – review & editing. **Yu Liu:** Conceptualization, Project administration, Writing – review & editing. **Donghao Li:** Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

The project was sponsored by the National Key Research and Development Program of China (2020YFB1902102). Thanks to experts from China Institute of Atomic Energy and Shanghai Nuclear Engineering Research and Design Institute for their inspiration and help.

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